COVERINGS BECOME SURJECTIONS AFTER SHEAFIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Surjections of sheaves as epimorphisms. Same chapter, Lemma lemma-mono-epi-sheaves. Reduces the assertion to the epimorphism property of the sheafified map.
- Sections of a representable sheaf. Same chapter, the identification of morphisms from a representable sheaf with sections of the target. Turns morphisms from the representable sheaf into sections so that the sheaf property can be applied.

Source reference. Original lemma 00WN: sources/stacks-sites/pinned-original-source.tex, source lines 2642–2642. Navigation pointer invoked at proof line 2642; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. If $\{U_i \to U\}_{i \in I}$ is a covering of the site $\mathcal{C}$, then the morphism of presheaves of sets*

$$\coprod_{i \in I} h_{U_i} \to h_U$$

becomes surjective after sheafification.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. By Lemma 00WN above we have to show that $\coprod_{i \in I} h_{U_i}^\# \to h_U^\#$ is an epimorphism. Let $\mathcal{F}$ be a sheaf of sets. A morphism $h_U^\# \to \mathcal{F}$ corresponds to a section $s \in \mathcal{F}(U)$. Hence the injectivity of $\mathrm{Mor}(h_U^\#, \mathcal{F}) \to \prod_i \mathrm{Mor}(h_{U_i}^\#, \mathcal{F})$ follows directly from the sheaf property of $\mathcal{F}$. $\square$